

Red Cabbage *pH Indicator*

Questions:

1. What is pH?
2. What is meant by 'acidic' and/or 'alkaline'?
3. What is an indicator?
 - a. Why does red cabbage work as an indicator?

Equipment:

- Red cabbage (duh!)
- Pestle and mortar.
- Filter paper and funnel..
- Spotting tiles.
- Solutions (there are a variety)

Safety: Please wear safety glasses, protect those eyes ;)

Method:

1. Begin by making a red cabbage pH indicator solution. Grind red cabbage in pestle and water and blend it with water. Filter the mixture to obtain a purple liquid.
2. Pour the red cabbage indicator solution into several small containers or clear plastic cups.
3. Label each cup with a number or a letter to represent different substances.
4. Prepare different solutions to test with the indicator. For example.
5. Add small amounts of each substance to the corresponding cups with the red cabbage indicator solution.
6. Observe the colour changes in the indicator solution. Red cabbage indicator changes colour based on the acidity or alkalinity of the substances. You can create a pH scale and compare the colours observed in each cup to identify whether the substance is acidic, neutral, or basic.

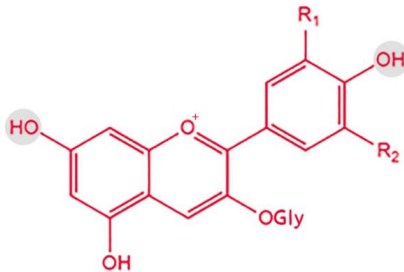


How does it work!?

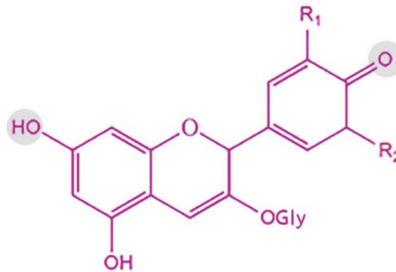
- ★ **What is pH?** - A measure of the acidity or alkalinity of a substance.
- ★ **What is an indicator?** - Weak acids which act as natural dyes and give a rough idea of what the pH of a solution is.
- ★ **What is meant by 'acidic' and 'alkaline'?** - Alkaline is a pH bigger than 7 and acidic is a pH less than 7

Red Cabbage:

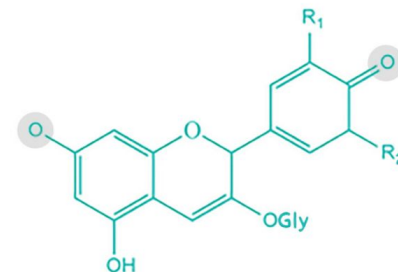
- ★ Red cabbage contains molecules called anthocyanins they are responsible for the cabbage's red colour.
- ★ The colour of these molecules depends on pH.
- ★ In acidic and alkaline solutions these molecules undergo structural changes which changes their colour.



ACIDIC



NEUTRAL



BASIC i.e alkaline

MAKING AN INDICATOR FROM RED CABBAGE

The compounds that give red cabbage its colour can be extracted and used as a pH indicator solution. Here we look at the method and the colours!

MAKING THE INDICATOR



1

ROUGHLY CHOP
THE CABBAGE



2

BOIL FOR A
FEW MINUTES



3

STRAIN AND
LET COOL

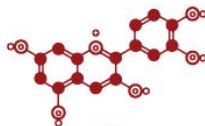


4

USE AS AN
INDICATOR!



← ACIDIC ————— pH ————— ALKALINE →



RED (pH <3)



VIOLET (pH 4-7)



BLUE (pH 7-8)



YELLOW GREEN (AT pH >8)

Hydrogens on carbon atoms implied; each carbon has 4 bonds.

The red cabbage extract can be used to determine whether substances are acidic or alkaline. The structures of the anthocyanin pigments which give the red cabbage its colour are subtly changed at varying pH. These different structures give a range of colours.

